JABBA: Just Another Bayesian Biomass Assessment

Nicola Walker | Principal Fisheries Stock Assessor



Centre for Environment Fisheries & Aquaculture Science

Cefas

Introduction

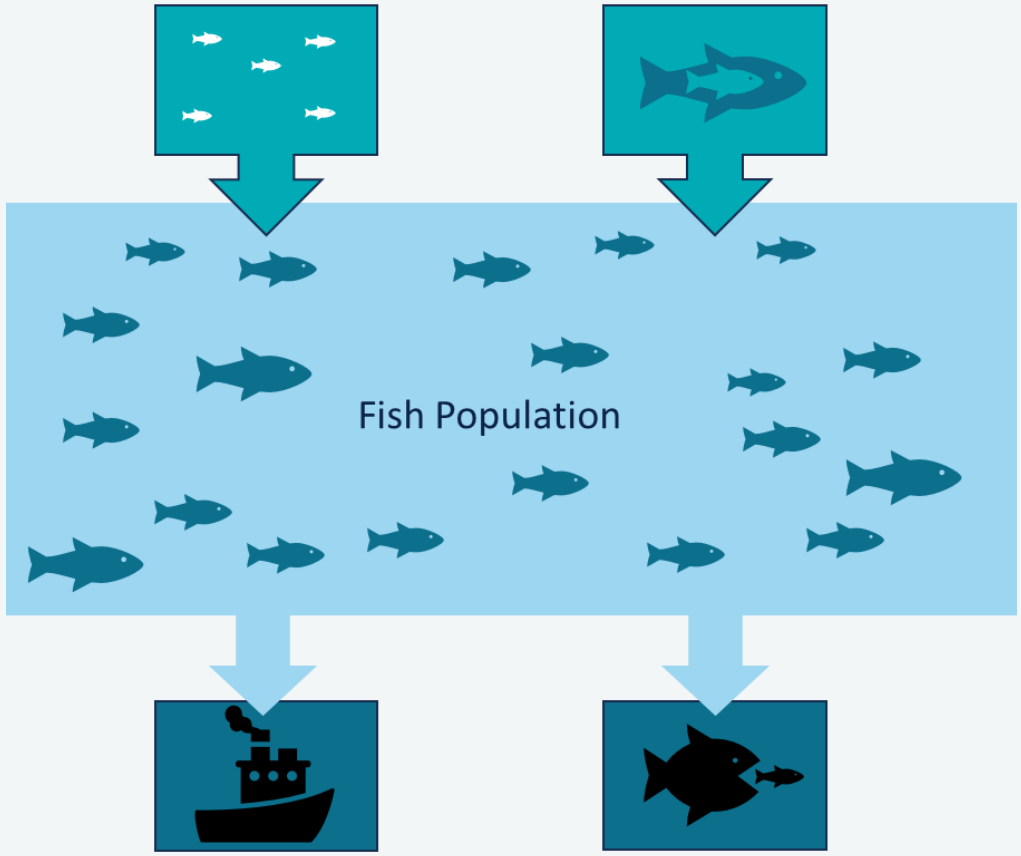
What is JABBA*?

- JABBA is a Bayesian state-space surplus production model (SPM) used for stock assessment
- SPMs group all individuals in a stock (no age or size structuring)
- SPiCT is another example of an SPM
- JABBA has been widely applied in stock assessments of sharks, tuna and billfishes around the world.

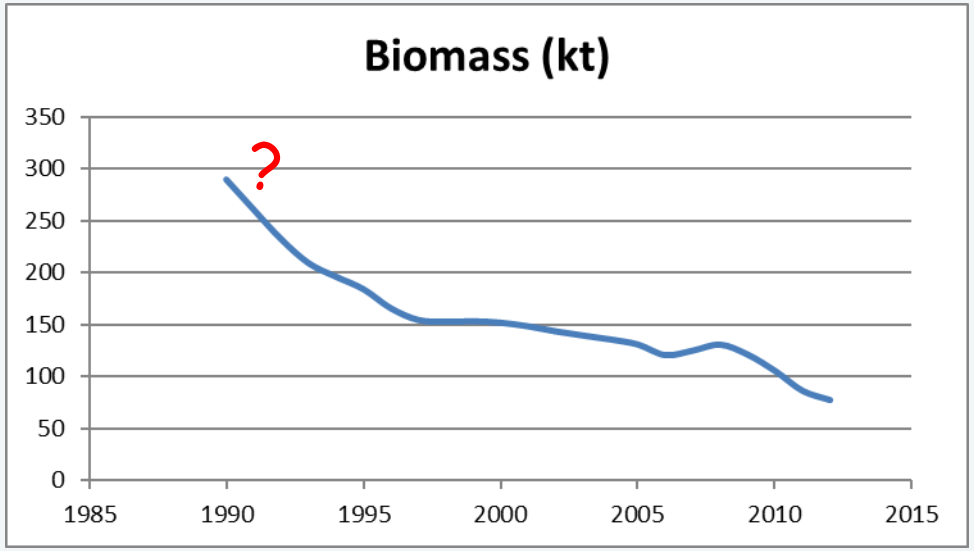
*I am NOT an expert in JABBA

Surplus production models

Population dynamics

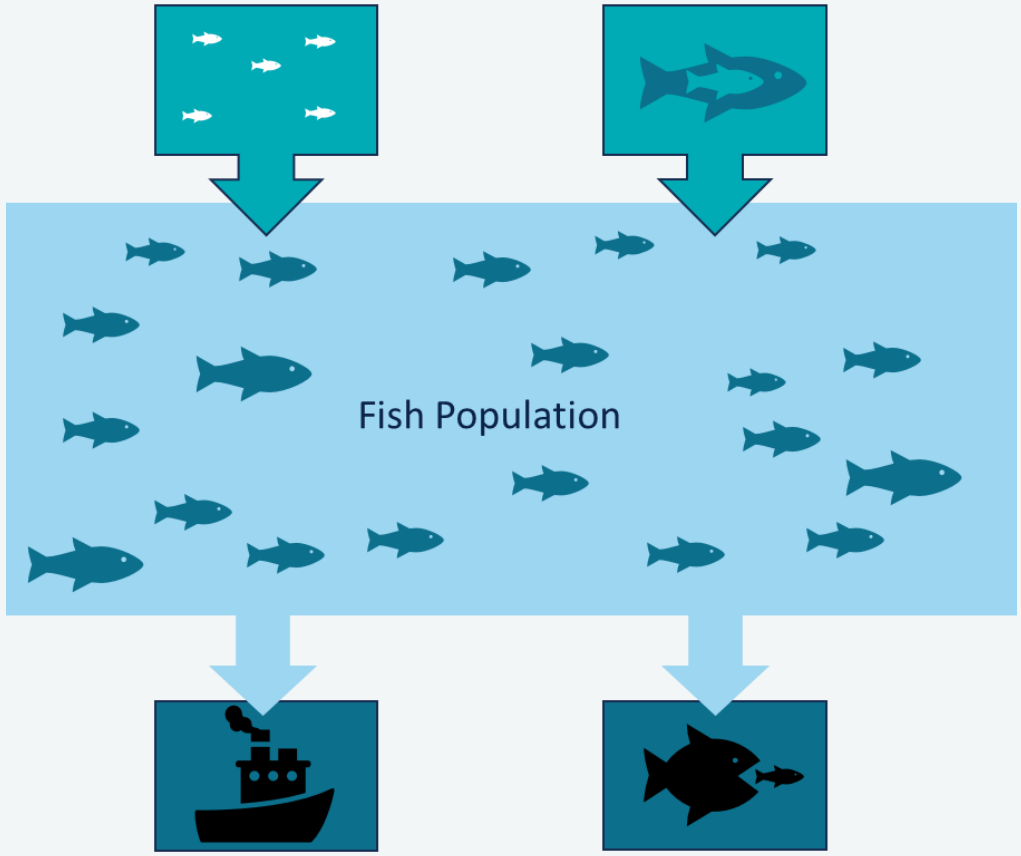


$$B_y = B_{y-1} + rB_{y-1}\left(1 - \frac{B_{y-1}}{K}\right) - C_{y-1}$$

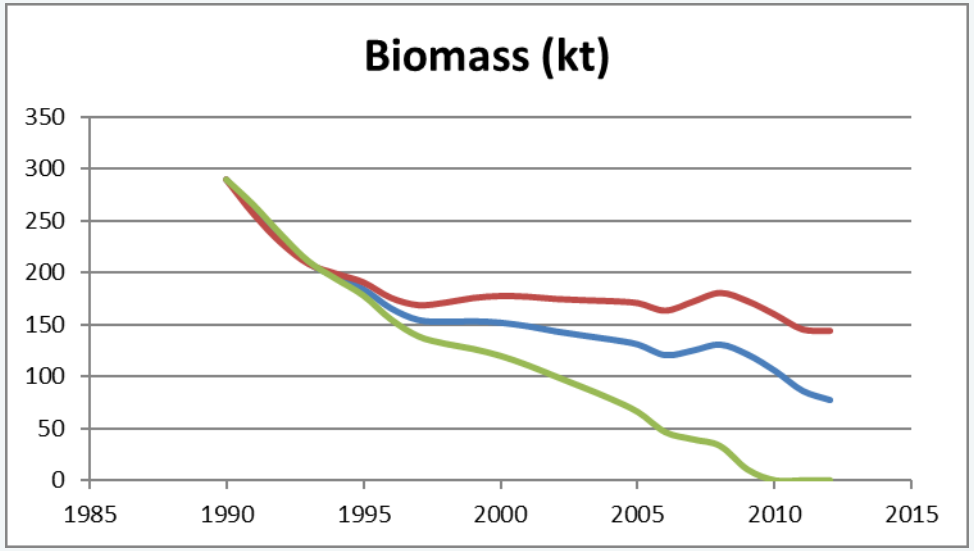


Surplus production models

Population dynamics



$$B_y = B_{y-1} + r B_{y-1}\left(1 - \frac{B_{y-1}}{K}\right) - C_{y-1}$$



Surplus production models

Observations & parameter estimation

B0	290
R	0.3
K	251
q	



Year	B	Catch	Effort	CPUE	$CPUE_{pred}$
1990	290	15.9	0.23	61.9	
1991	260	25.7	0.41	79.0	
1992	232	28.5	0.51	55.6	
1993	209	23.7	0.47	44.6	

$$I_y = \overset{?}{q} B_y$$

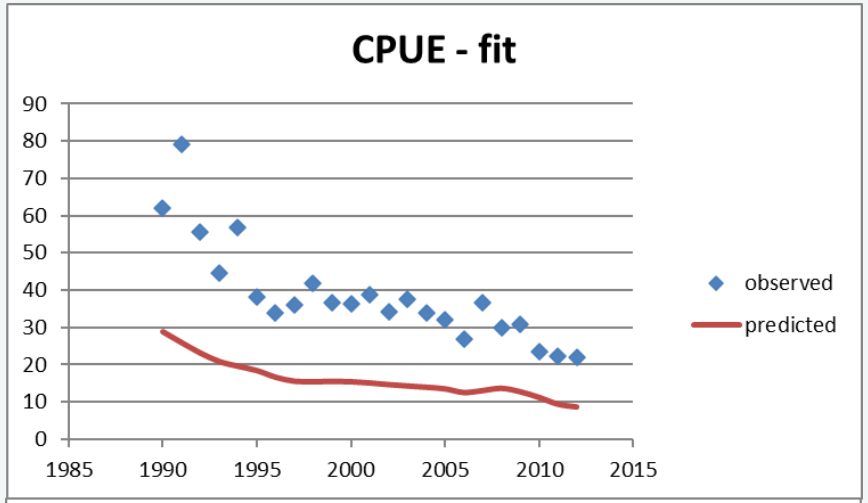
Surplus production models

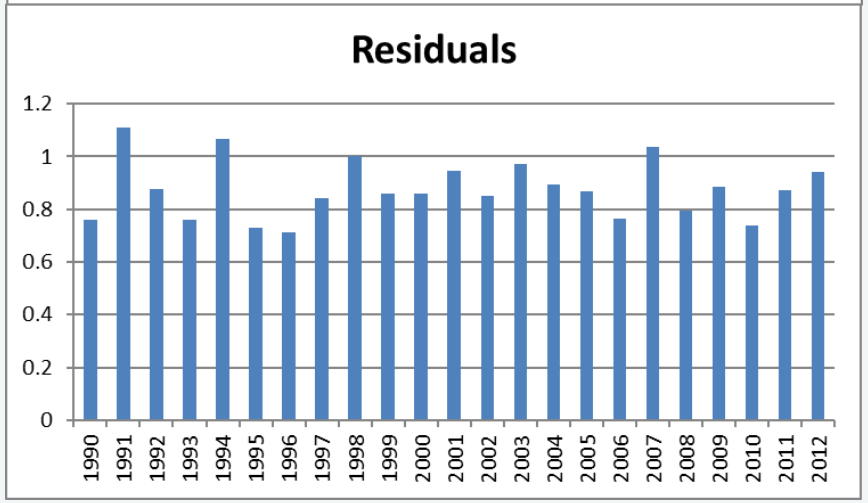
Observations & parameter estimation

B0	290
R	0.3
K	251
q	0.1



Year	B	Catch	Effort	CPUE	$CPUE_{pred}$
1990	290	15.9	0.23	61.9	29
1991	260	25.7	0.41	79.0	26
1992	232	28.5	0.51	55.6	23
1993	209	23.7	0.47	44.6	21





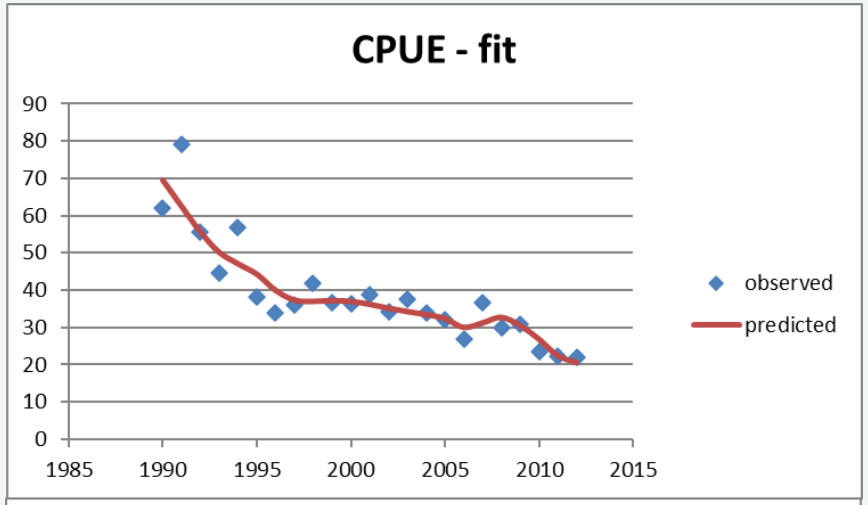
Surplus production models

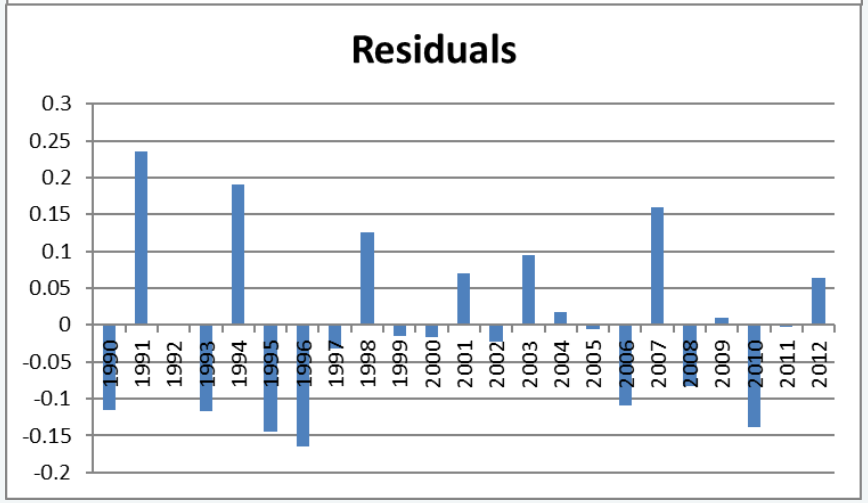
Observations & parameter estimation

B0	290
R	0.3
K	251
q	0.24



Year	B	Catch	Effort	CPUE	$CPUE_{pred}$
1990	290	15.9	0.23	61.9	69.4
1991	260	25.7	0.41	79.0	62.4
1992	232	28.5	0.51	55.6	55.6
1993	209	23.7	0.47	44.6	50.1



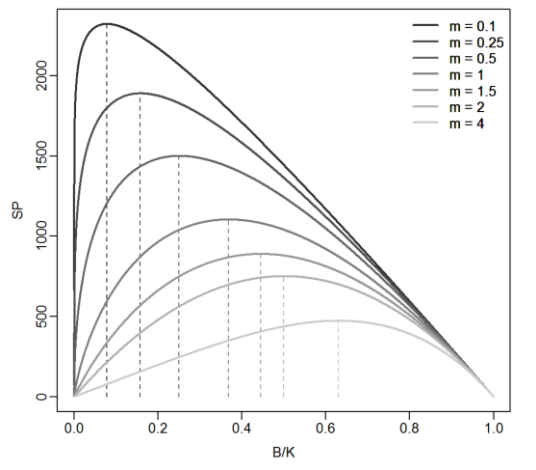


JABBA

Just a little more complicated

Process error – the model is a simplification of the real world!

Population dynamics: $B_y = \left(B_{y-1} + \frac{r}{m-1} B_{y-1}\left(1 - \left(\frac{B_{y-1}}{K}\right)^{m-1}\right) - \sum_f C_{f,y-1}\right) e^{\eta_y}$



Observations: $I_{i,y} = q_i B_y e^{\varepsilon_{y,i}}$ → Observation error – data are noisy!

You can have more than one

Parameter estimation: $\theta = \{K, r, \varphi, \sigma_\eta^2, q_i, \sigma_{\varepsilon,y,i}^2\}$

B_0/K

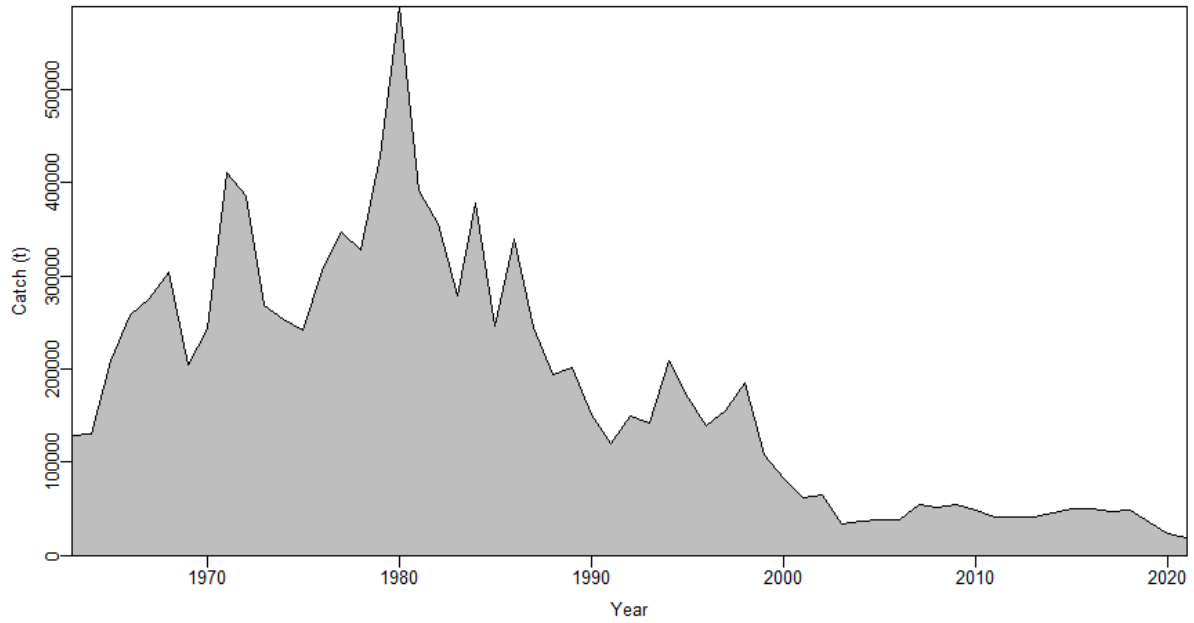
Can be fixed or estimated

Split into three components that can be switched on or off in any combination

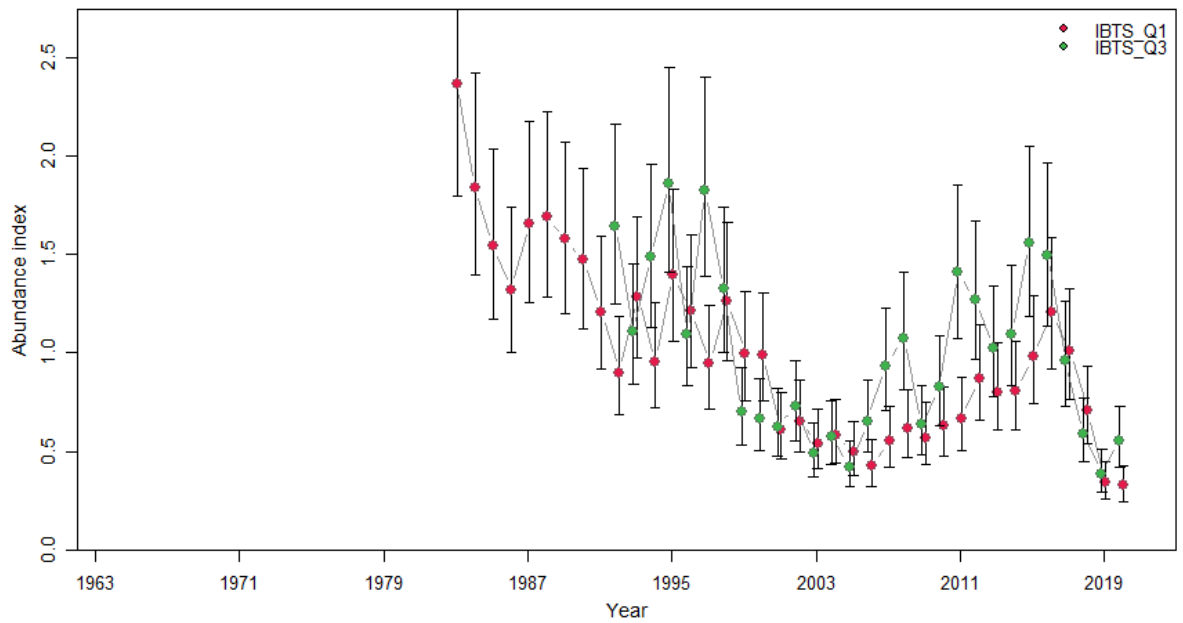
JABBA demo: North Sea* cod

Input data

Catch



Indices**



+ priors

Can also include associated standard error estimates (I didn't do this)

* The old stock definition

** Processed using VAST

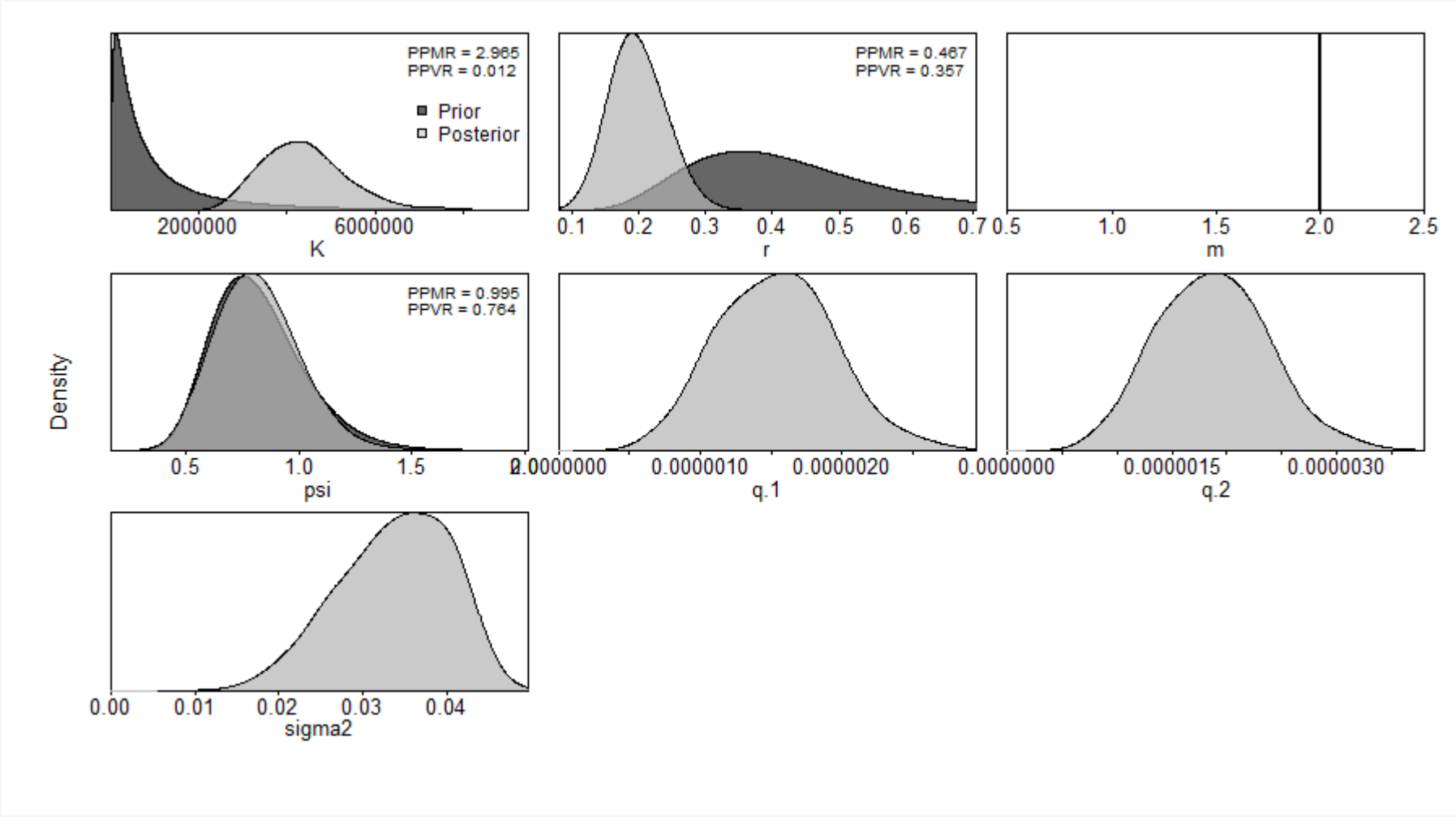
JABBA demo: North Sea cod

My modelling process

Run	model	BmsyK	catch.cv	catch.error	sigma.proc	proc.dev.all	igamma	shared_q	sigma.est	shared_var	fixed.obsE	Convergence	SDNR	RMSE	DIC	Runs test	PPC_1	PPC_2
Production model																		
3a	schaefer		0.1	random	TRUE	TRUE	informative	NA	TRUE	NA	FALSE	Yes	0.57	7.4	-15.8	Yes	0.495	NA
3b	fox		0.1	random	TRUE	TRUE	informative	NA	TRUE	NA	FALSE	No	0.44	4.7	49.8			
3c	pella	0.4	0.1	random	TRUE	TRUE	informative	NA	TRUE	NA	FALSE	Yes	0.44	4.4	106.9			
3d	pella_m	0.4	0.1	random	TRUE	TRUE	informative	NA	TRUE	NA	FALSE	No	0.6	7.9	-123.3			
Add IBTS_Q3																		
4	schaefer		0.1	random	TRUE	TRUE	informative	No	TRUE	No	FALSE	Yes	0.8	18	-378.4	No	0.492	0.511
Process error																		
5a	schaefer		0.1	random	0.1	TRUE	informative	No	TRUE	No	FALSE	Yes	0.9	22	-450.3	Yes	0.5	0.492
5b	schaefer		0.1	random	0.1	FALSE	informative	No	TRUE	No	FALSE	Yes	0.97	45.2	3383183.1	No	0.498	0.501
5c	schaefer		0.1	random	TRUE	FALSE	informative	No	TRUE	No	FALSE	Yes	0.8	22	127682	No	0.499	0.496
5e	schaefer		0.1	random	TRUE	FALSE	uninformative	No	TRUE	No	FALSE	Yes	0.79	21.4	1647737.2	No	0.503	0.494
5f	schaefer		0.1	random	TRUE	TRUE	uninformative	No	TRUE	No	FALSE	Yes	0.79	18	-416	No	0.506	0.495
Observation error																		
6a	schaefer		0.1	random	TRUE	TRUE	uninformative	No	**FALSE**	No	FALSE	Yes	142.24	15.2	3699344.3	No	**0**	**0**
6b	schaefer		0.1	random	TRUE	TRUE	uninformative	No	**FALSE**	No	0.01	No	14.22	15.2	12455.4	No	**0**	**0**
6c	schaefer		0.1	random	TRUE	TRUE	uninformative	No	**FALSE**	No	0.1	Yes	1.48	15.7	-470.3	No	**0.001**	**0**
6d	schaefer		0.1	random	TRUE	TRUE	uninformative	No	TRUE	No	0.1	No	0.81	18	-467.4	No	0.501	0.484
6e	schaefer		0.1	random	TRUE	TRUE	uninformative	Yes	TRUE	No	FALSE	No	0.82	21.4	-359.1	Yes	0.511	0.5
6f	schaefer		0.1	random	TRUE	TRUE	uninformative	Yes	TRUE	**Yes**	FALSE	No	0.87	20.4	-463.3	Yes	**0.677**	**0.305**
6g	schaefer		0.1	random	TRUE	TRUE	uninformative	No	TRUE	**Yes**	FALSE	Yes	0.85	17.8	-433.5	No	**0.687**	**0.304**
6h	schaefer		0.1	random	TRUE	TRUE	uninformative	No	TRUE	No	0.15	No	0.81	17.7	-489.4	Yes	0.606	0.456
6i	schaefer		0.1	random	TRUE	TRUE	uninformative	No	TRUE	No	0.16	Yes	0.82	17.7	-488.6	No	0.637	0.442
6i	schaefer		0.1	random	TRUE	TRUE	uninformative	No	TRUE	No	0.17	Yes	0.83	17.8	-488.6	No	0.646	0.417
6i	schaefer		0.1	random	TRUE	TRUE	uninformative	No	TRUE	No	0.18	No	0.83	17.9	-487.8	No	0.693	0.425
6i	schaefer		0.1	random	TRUE	TRUE	uninformative	No	TRUE	No	0.19	Yes	0.82	18.1	-487.3	No	0.744	0.45
6i	schaefer		0.1	random	TRUE	TRUE	uninformative	No	TRUE	No	0.14	Yes	0.83	17.9	-486.8	Yes	0.57	0.463
Catch error																		
7a	schaefer		0	random	TRUE	TRUE	uninformative	No	TRUE	No	0.14	No	0.82	17.9	-487.2	Yes	0.579	0.459
7b	schaefer		0.1	under	TRUE	TRUE	uninformative	No	TRUE	No	0.14	No	0.83	18	-487	Yes	0.571	0.461
7c	**schaefer**		**0.15**	**random**	**TRUE**	**TRUE**	**uninformative**	**No**	**TRUE**	**No**	**0.14**	**Yes**	**0.83**	**17.9**	**-490.1**	**Yes**	**0.559**	**0.448**
Production model - again																		
8a	pella_m	0.4	0.15	random	TRUE	TRUE	uninformative	No	TRUE	No	0.14	No	0.83	17.9	-485.9	Yes	0.559	0.46
8b	pella	0.53	0.15	random	TRUE	TRUE	uninformative	No	TRUE	No	0.14	Yes	0.83	17.9	-488.9	Yes	0.562	0.453

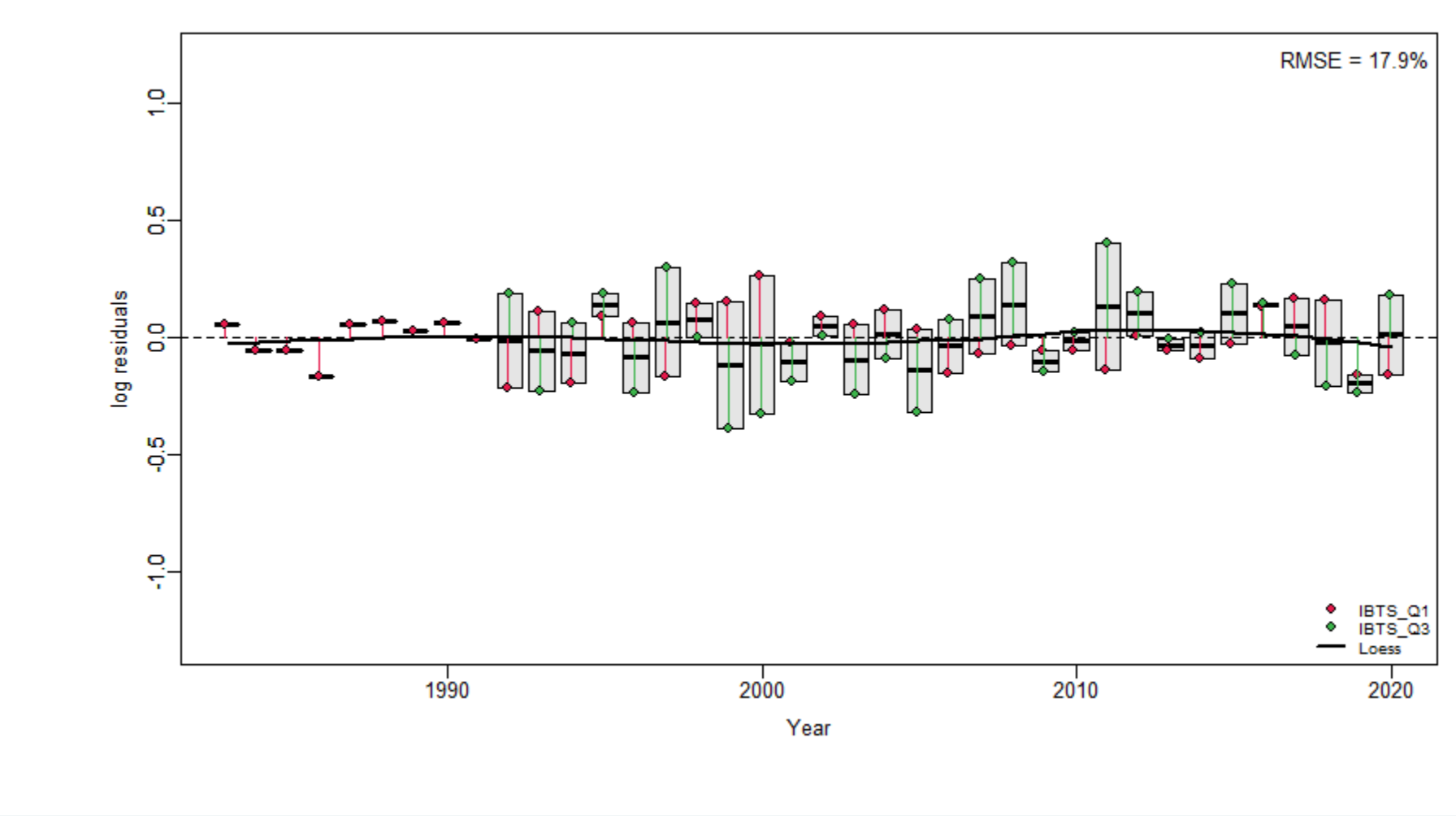
JABBA demo: North Sea cod

Some (not all) diagnostic plots



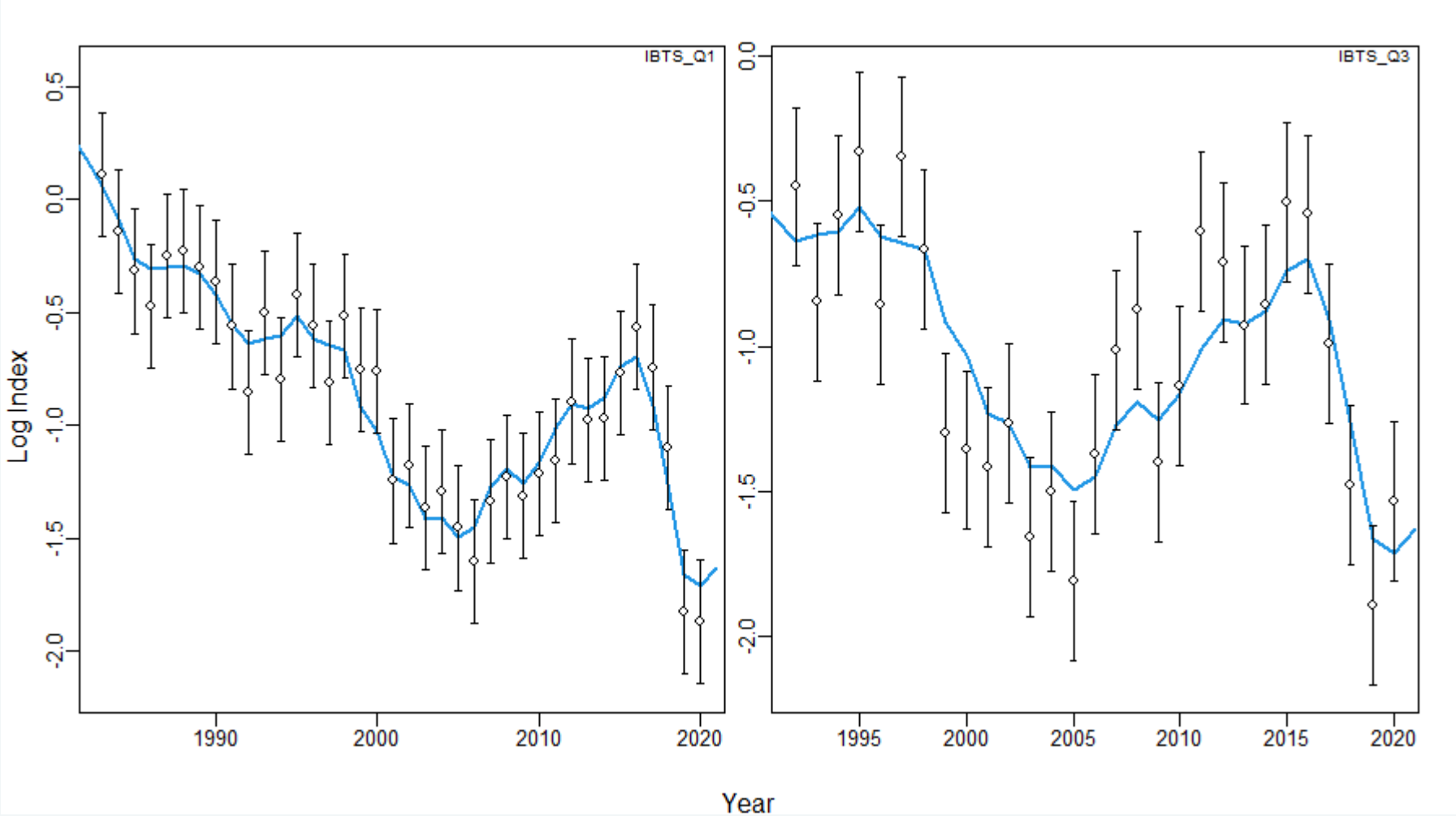
JABBA demo: North Sea cod

Some (not all) diagnostic plots



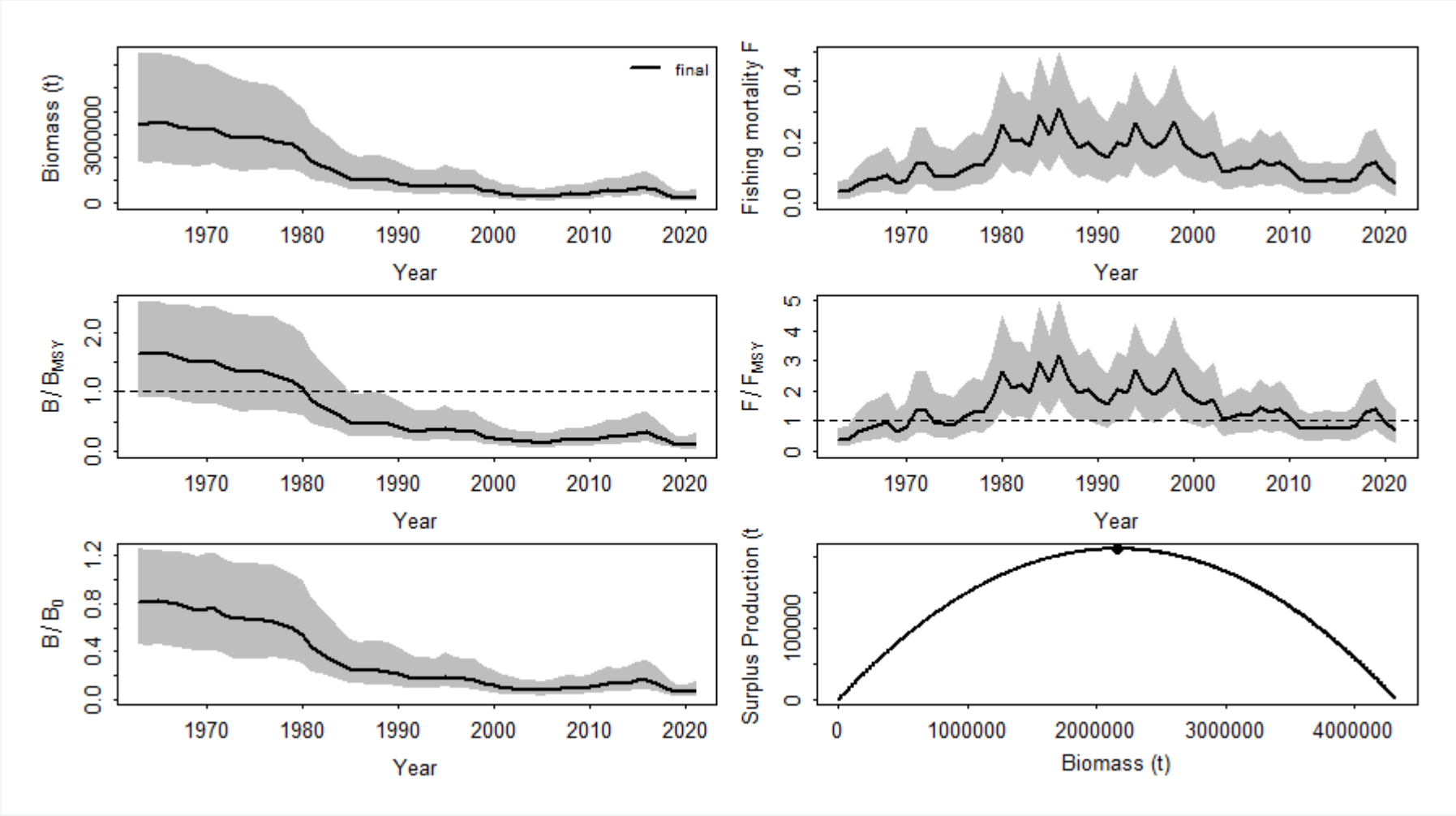
JABBA demo: North Sea cod

Some (not all) diagnostic plots



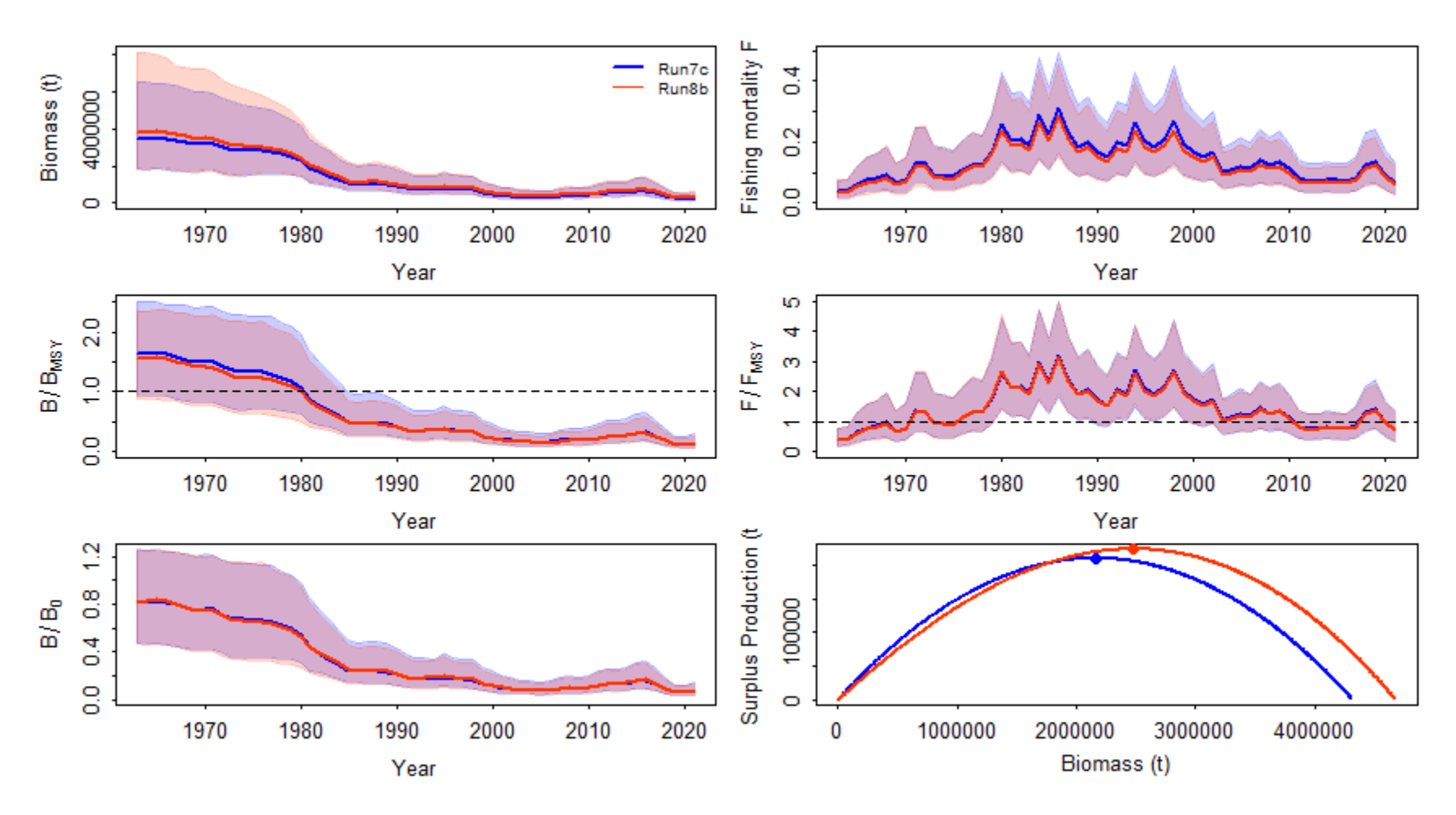
JABBA demo: North Sea cod

Outputs



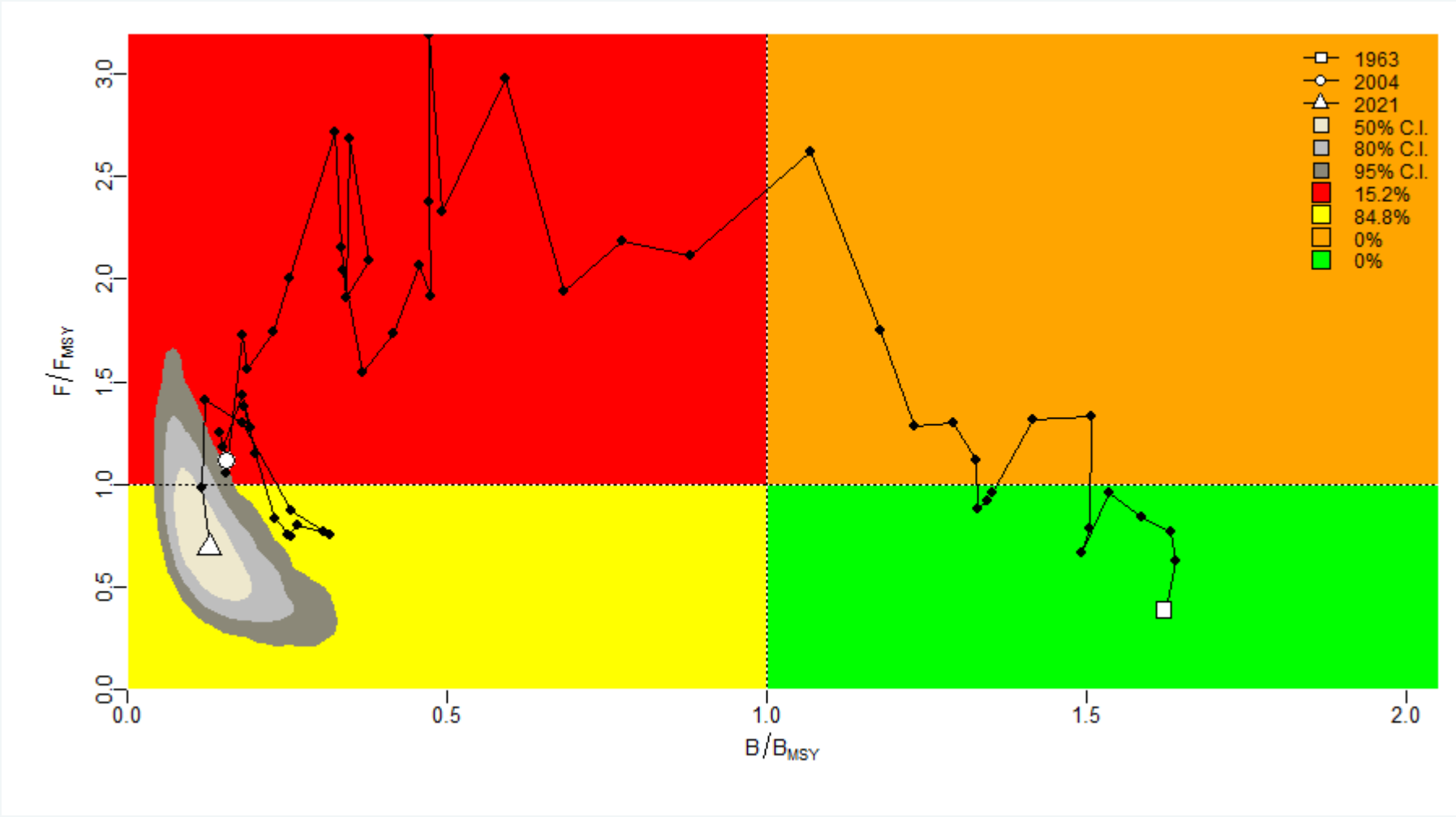
JABBA demo: North Sea cod

Outputs



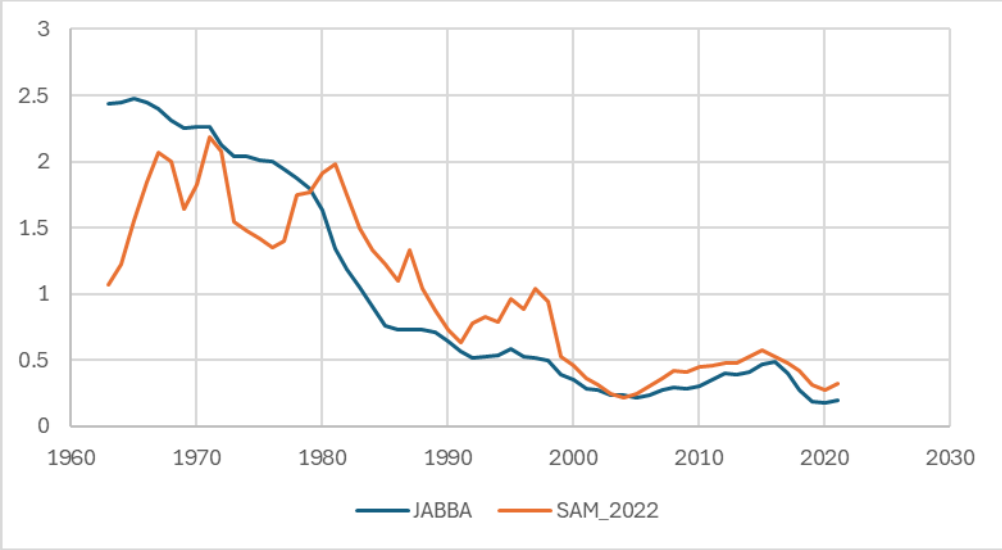
JABBA demo: North Sea cod

Outputs



JABBA demo: North Sea* cod

JABBA vs SAM



- Agreement on stock and exploitation status between runs. And pretty much where I expect them to be
- Big differences in absolute TSB but the trends are (more or less) there
- Shares SAMs retro pattern (not shown)

JABBA						SAM			
	mu	lci	uci			TSB (tonnes)	Low	High	
1963	3428734	1849460	6507378	2.435327		314632	261652	378339	1.068719
1964	3451505	1796211	6508278	2.4515		360042	307700	421287	1.222964
1965	3490296	1850773	6487802	2.479053		457846	397012	528002	1.555177
1966	3449323	1797332	6442453	2.44995		543430	475595	620940	1.845883
1967	3370234	1739766	6358007	2.393776		608427	534342	692784	2.06666

JABBA

Discussion

- Does anyone else use or have experience with JABBA?

- What are the differences between JABBA and SPiCT?
  - JABBA is Bayesian and SPiCT based on maximum likelihood?
  - SPiCT models *F* as a process and JABBA takes the catch as is (with recent option to include some prespecified error)?
  - SPiCT is in continuous time (what does this mean)?

- Are there reasons one is used over the other?
  - ICES vs RFMOs?
  - Expertise?

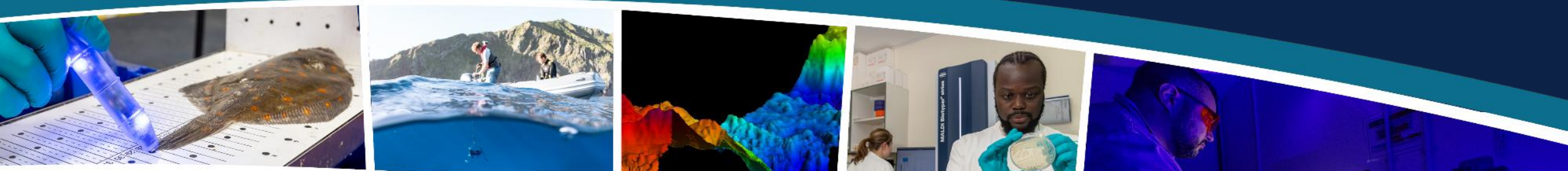
Thank you for listening

Where to find JABBA:

[jabbamodel/JABBA: Just Another Bayesian Biomass Assessment (github.com)]

To learn about Cefas scan the QR code or visit
linktr.ee/CefasGovUK

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